P12 4/4

Lecture 18: Section 2.7
Nonlinear Inequalities

Polynomial inequalities

Critical (key) numbers

Rational inequalities

Lecture 18: Section 2.7
Nonlinear Inequalities

To Solve a Polynomial Inequality

1. Write the inequality with 0 on the right hand side to obtain one of the following:

$$f(x) > 0 \quad f(x) \geq 0 \quad f(x) < 0 \quad f(x) \leq 0$$

Factor

2. Find all real zeros of the polynomial $f(x)$. These zeros are the **critical numbers** of $f(x)$.
or Key numbers

3. Use the critical numbers to determine the test intervals.

4. Choose a number in each interval and evaluate f at the number.

a) If the function value is positive, then $f(x) > 0$ for all numbers x in the interval.

b) If the function value is negative, then $f(x) < 0$ for all numbers x in the interval.

5. Include endpoints if $f(x) \geq 0$ or if $f(x) \leq 0$.

FACT: A polynomial changes signs only at its zeros. Between two zeros, a polynomial must be entirely positive or entirely negative.

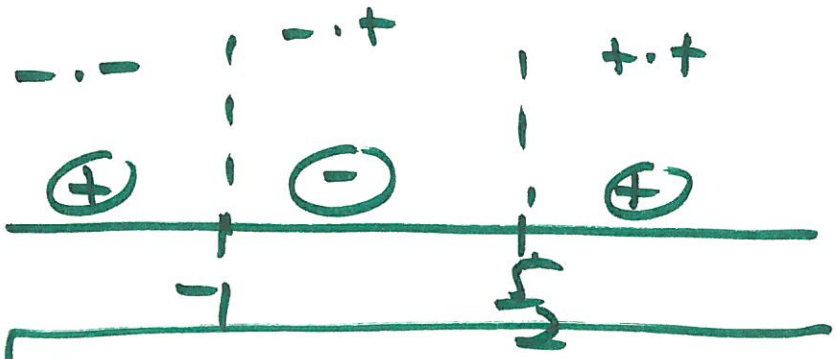
<u>ex.</u> Solve the inequalities: $2x^2 - 3x > 5$. Write your answer in interval notation.

① $2x^2 - 3x - 5 > 0$

② $(2x-5)(x+1) > 0$

critical #s $x = \frac{5}{2}, -1$

③



④ $\boxed{(-\infty, -1) \cup (\frac{5}{2}, \infty)}$

* Don't ÷ or × by both sides by a variable since it makes it harder. You would have to do cases.

It might be easier to use the factored form of the polynomial to determine the signs of the polynomial in each interval.

<u>ex.</u> Solve the inequality: $x^4 \leq 8x$.

$$\frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

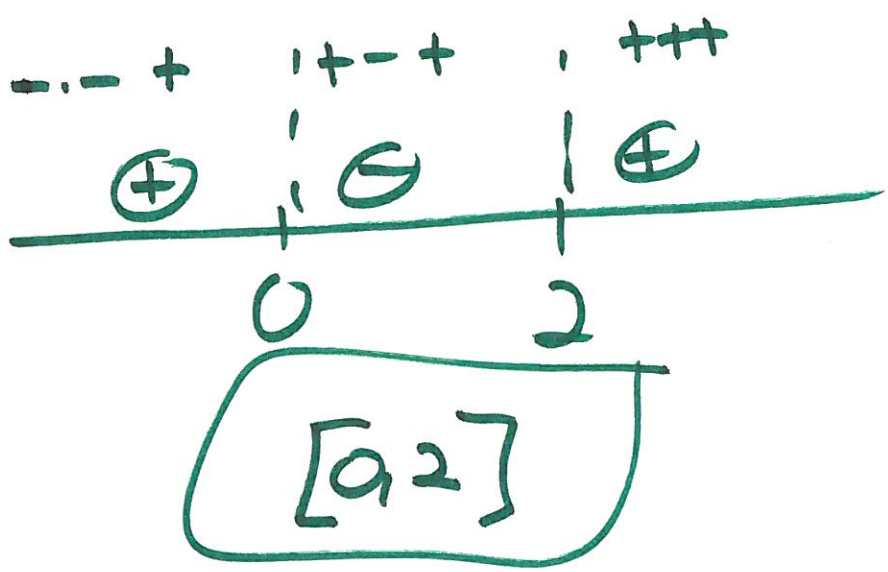
$x^4 - 8x \leq 0$

$x(x^3-8) \leq 0$

$x(x-2)(x^2+2x+4) \leq 0$

$b^2-4ac \rightarrow 4-4(1)4 = \text{neg.}$ thus no real # solutions

critical #s $x = 0, 2$



$\boxed{[0,2]}$

To Solve a Rational Inequality

1. Write the rational inequality with 0 on the right side. Write the left side as a **single** fraction. Note the domain.

2. Find all zeros of numerator and denominator. These zeros are the **critical numbers** of a rational inequality.

3. Use the critical numbers to determine the test intervals.

4. Solve as before, including endpoints **in the domain** if they are solutions.

FACT: The value of a rational expression can change signs at its zeros and its undefined values. These two types of numbers make up the **critical numbers** of a rational inequality.

① zero on one side ✓

② Single fraction on other side ✓

③

ex. Solve the inequality: $\dfrac{(x+1)(4-x)}{(x-2)^2} > 0$

critical #s: $x = -1, 4, 2$

$\dfrac{-+}{+}$ | $\dfrac{++}{+}$ | $\dfrac{++}{+}$ | $\dfrac{+-}{+}$

$-$ | $+$ | $+$ | $-$

-1 $\quad 2 \quad 4$

$$\boxed{(-1, 2) \cup (2, 4)}$$

$$\cancel{(-1, 4)}$$

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ex. $\dfrac{(x+1)(4-x)}{(x-2)^2} \geq 0$

$$\boxed{[-1, 2) \cup (2, 4]}$$

① Zero on one side

$$\frac{6}{2x-5} - 2 \le 0$$

ex. Solve the inequality: $\dfrac{6}{2x-5} \le 2$

② Get single fraction on left side

$$\frac{6}{2x-5} - 2\left(\frac{2x-5}{2x-5}\right) \le 0$$

③ critical #s

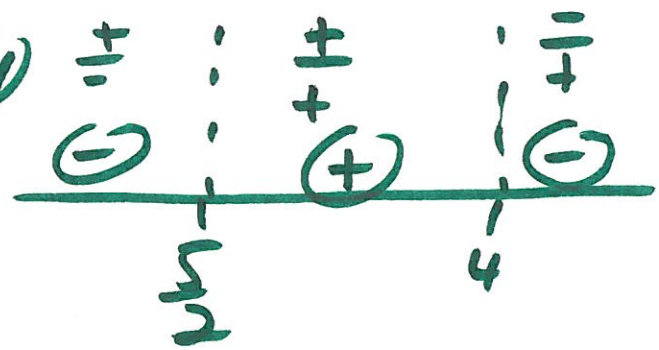
$x = 4, \frac{5}{2}$

$$\frac{6-2(2x-5)}{2x-5} \le 0$$

$$\frac{6-4x+10}{2x-5} \le 0$$

$$\frac{16-4x}{2x-5} \le 0$$

④



$$\left(-\infty, \tfrac{5}{2}\right) \cup [4, \infty)$$

NOTE: Do not clear fractions in an inequality when there is a variable in the denominator.

$$\cancel{(2x-5)\left(\frac{6}{2x-5}\right) \le (2)(2x-5)}$$

$$\cancel{6 \le 4x-10}$$

$$\cancel{16 \le 4x}$$

$$\cancel{4 \le x}$$

$$\cancel{[4, \infty)}$$

(1) Get 0 on one side.

ex. Solve the inequality $\frac{3}{x} + \frac{4}{x-2} \le \frac{3}{x} - \frac{3}{x}$

$$\frac{4}{x-2} - \frac{3}{x} \le 0$$

(2) Get single fraction.

$$\left(\frac{4}{x-2}\right)\frac{x}{x} - \frac{3}{x}\left(\frac{x-2}{x-2}\right) \le 0$$

$$\frac{4x - 3(x-2)}{x(x-2)} \le 0$$

$$\frac{4x - 3x + 6}{x(x-2)} \le 0$$

$$\frac{x+6}{x(x-2)} \le 0$$

(3) critical #s

$x = -6$

$x = 0$

$x = 2$

(4)

	-6		0 und		2 und	
$\frac{-}{--}$		$\frac{+}{--}$		$\frac{+}{+-}$		$\frac{+}{++}$
$-$		$+$		$-$		$+$

$$\boxed{(-\infty, -6] \cup (0, 2)}$$

not in domain, thus (,) and not [,].

Applications

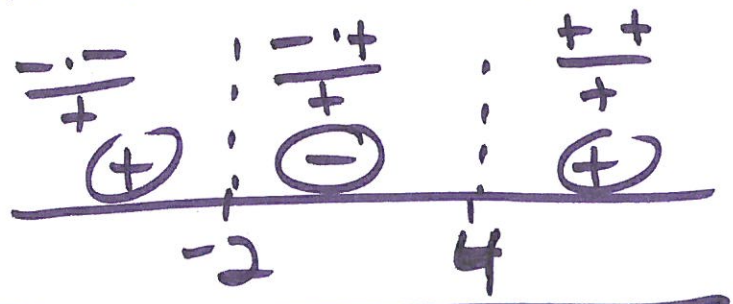
<u>ex.</u> Find the domain of the function:

$$f(x) = \sqrt{\frac{x^2 - 2x - 8}{x^2 + 1}}$$

$$\frac{x^2-2x-8}{x^2+1} \geq 0$$

$$\frac{(x-4)(x+2)}{x^2+1} \geq 0$$

critical #s $x = 4, -2$



$$\boxed{(-\infty, -2] \cup [4, \infty)}$$

If an object is subject only to the force of gravity, then the object's height, $h(t)$, is given by the formula

$$h(t) = -16t^2 + v_0 t + h_0$$

where h_0 is the **initial height** in feet and v_0 is the **initial velocity** in feet per second.

NOTE: If the object is moving upward at $t = 0$, then $v_0 > 0$, whereas if the object is moving downward at $t = 0$, then $v_0 < 0$.

<u>ex.</u> A projectile is fired straight upward from the ground level with an initial velocity of 384 feet per second. During what time period will its height exceed 2048 feet?

$h(t) = -16t^2 + 384t + 0$

$h(t) = -16t^2 + 384t$

$-16t^2 + 384t > 2048$

$-16t^2 + 384t - 2048 > 0$

$-16(t^2 - 24t + 128) > 0$

$-16(t-8)(t-16) > 0$

$(t-8)(t-16) < 0$

$- - - \;|\; - + - \;|\; - + +$

$\ominus \;|\; \oplus \;|\; \ominus$

$8 \quad 16$

$(8, 16)$ sec.

$8 < t < 16$ sec

Practice.

1) Solve the inequality: $\dfrac{1}{x^2+8x+12} \geq \dfrac{1}{x}$

2) The demand function for a product is $p(x) = -x + 10,\ 0 \leq x \leq 10$. The cost function for the product is $C(x) = 2x + 12$. Find all values of x for which the company will earn a profit. Express your answer in interval notation.

Answer. 1) $(-\infty, -6) \cup [-4, -3] \cup (-2, 0)$ 2) $(2, 6)$